1 Purpose and Motivations

1. Basic equation: Start with general Navier-Stokes; specialize to incompressible, Newtonian fluids; how many equations needed to solve uniquely; systematically derive boundary conditions at free surface using exact surface evolution equation, with special attention to surface tension (derive ST pressure/force); specialize to inviscid fluids; irrotational flow and velocity potential (Laplace eq.); Bernoulli equation for irrotational flow

2. Small oscillations of an incompressibe, inviscid, irrotational (introduce I3) drop: Laplace equation (solve in spherical coordinates); surface function and deviation; integral constraints on deviation; linearization about a static equilibrium (hence, irrotational) — address kinematic SBC and Bernoulli equation evaluated at the surface; combine kinematic and surface Bernoulli, substitute normal mode and derive dispersion relation, derive normal-mode frequency, show capillary frequency scaling

3. Total energy of the oscillating drop: KE; Surface energy (Reynold's transport); total mechanical energy (virial)

4. Viscous Dissipation: general dissipation integral (isotropic); specialize to Newtonian, incompressible; Spectator limit of dissipation rate and Oh

5. Appendix A (Reynold's transport equation): Derivation using delta-function formalism; derivation using total derivative of integral

6. Virial theorem: Immediately specialize to incompressible fluid; tricky because of surface term and retention of second-order terms

The purpose of these notes is to provide a largely self-contained discussion of the dynamics and mechanical energy of an oscillating liquid drop and the rate at which the energy decreases due to viscous dissipation. This subject has a long history that extends back to Rayleigh, and much of the relevant material is spread throughout standard fluid mechanics text books, such as Lamb (1945), Landau & Lifshitz (1987), and Kundu & Cohen (2002); we have drawn liberally from these sources. However, it is worth pointing out that, to the best of our knowledge, these authors provide no explicit mathematical exposition on the potential energy stored in surface tension. An attempt is made here to fill this intellectual gap by giving a rather complete discussion of the surface energy and its rate of change.

Our presentation begins in §4.1 with highlights from the linear theory of an inviscid oscillating drop. These results allow us to obtain the total kinetic energy and surface energy in §§4.2 and 4.3, from which we obtain in §4.4 the drop's conserved total mechanical energy. In §5, we introduce viscosity into the discussion, first (§5.1) in a very general way when deriving the rate at which mechanical energy is converted to heat, and then (§5.2) using this rate and our inviscid results to obtain an approximate expression for the mean energy dissipation rate in the limit of small viscosity. Our physical discussion concludes with brief remarks on the interpretation of the dimensionless Ohnesorge number often encountered in the literature on oscillating, viscous drops. Finally, in the Appendix, we rigorously derive expressions for the time derivatives of volume and surface integrals with variable domains.

In all that follows, we assume that the drop is surrounded by vacuum and that the liquid density ρ, surface tension σ, and dynamic viscosity μ are constant. Moreover, although we consider viscous friction, we neglect internal thermal energy and heat transfer.

2 Fundamentals

Here we review the basic equations dictated by the requirements of mass and momentum conservation in fluid flow. While this discussion is rather elementary, and will likely be review for many readers, it serves another purpose by introducing several notational conventions used throughout these notes.

2.1 Mass Conservation

Imagine drawing a closed surface that surrounds a fixed volume of fluid. Denote the spatial domains occupied by the fixed volume and its bounding surface by V_0 and A_0, respectively. If ρ is the fluid density, the mass contained within V_0 is

$$\int_{V_0} dV \rho \ , \tag{1}$$

where dV is a differential volume element. A change in this contained mass is attributed to a net loss or gain of material through the surface A_0; this is the verbal statement of mass conservation. Let $d\boldsymbol{A} = dA\hat{\boldsymbol{n}}$ be a vector area element on A_0, where dA is the differential surface area, and $\hat{\boldsymbol{n}}$ is a unit vector perpendicular to the surface. Following the usual convention, we define the direction of $\hat{\boldsymbol{n}}$ such that fluid moving across the boundary in the direction of $\hat{\boldsymbol{n}}$ is *exiting* the volume in question (i.e., $\hat{\boldsymbol{n}}$ is the *outward* surface normal). If the fluid velocity is $\boldsymbol{u}$ in the neighborhood of some element of area on the boundary, the rate at which mass leaves V_0 through this patch of the surface is $\rho\boldsymbol{u} \cdot d\boldsymbol{A}$. When $\rho\boldsymbol{u} \cdot d\boldsymbol{A} > 0$, mass is being subtracted locally from V_0. The total rate at which mass exits V_0 is obtained by integrating $\rho\boldsymbol{u} \cdot d\boldsymbol{A}$ over the entire boundary A_0:

$$\partial_t \int_{V_0} dV \rho = -\int_{A_0} d\boldsymbol{A} \cdot (\rho\boldsymbol{u}) \ , \tag{2}$$

where we have used the shorthand $\partial_t \equiv \partial/\partial t$ for the time derivative at fixed spatial location. An application of the divergence theorem converts the above surface integral into an integral over the interior volume:

$$\int_{A_0} d\boldsymbol{A} \cdot (\rho\boldsymbol{u}) = \int_{V_0} dV \boldsymbol{\nabla} \cdot (\rho\boldsymbol{u}) \ . \tag{3}$$

Because V_0 is a fixed spatial domain[1], the time derivative in eq. (2) can be brought inside the integral, and thus

$$\int_{V_0} dV [\partial_t \rho + \boldsymbol{\nabla} \cdot (\rho\boldsymbol{u})] = 0 \ . \tag{4}$$

Since V_0 is an arbitrary volume, the above integrand must vanish everywhere within the fluid:

$$\partial_t \rho + \boldsymbol{\nabla} \cdot (\rho\boldsymbol{u}) = 0 \ . \tag{5}$$

This is the most common form of the differential equation for mass conservation.

An alternative form of eq. (5) is obtained if we expand the divergence, so that

$$(\partial_t + \boldsymbol{u} \cdot \boldsymbol{\nabla})\rho + \rho\boldsymbol{\nabla} \cdot \boldsymbol{u} \ . \tag{6}$$

For some function f that depends on time t and space $\boldsymbol{r}$, the total differential along an infinitesimal path (i.e., the directional derivative) is

$$df = dt\, \partial_t f + d\boldsymbol{r} \cdot \boldsymbol{\nabla} f \ . \tag{7}$$

Suppose that we choose to follow a particular tiny parcel of fluid that moves with velocity $\boldsymbol{u}$ at time t. This implies a constraint that relates the changes $d\boldsymbol{r}$ and dt; specifically, $d\boldsymbol{r} = \boldsymbol{u}\, dt$. In this special frame that co-moves with the fluid, we denote the rate of change of the function f by

$$D_t f = (\partial_t + \boldsymbol{u} \cdot \boldsymbol{\nabla}) f \ . \tag{8}$$

[1] The more general case of time variable domains will be addressed below.

This derivative is referred to by different names, including the *total derivative*, *material derivative*, *advective derivative*, and *Lagrangian derivative*, and is distinguished from the ordinary partial derivative (or *Eulerian derivative*), which is evaluated at a fixed coordinate. Given the definition of D/Dt, eq. (6) becomes

$$\frac{D_t\rho}{\rho} + \boldsymbol{\nabla}\cdot\boldsymbol{u} = 0\ . \tag{9}$$

This brings to light the physical interpretation of the velocity divergence $\boldsymbol{\nabla}\cdot\boldsymbol{u}$. According to eq. (9), $dt\,\boldsymbol{\nabla}\cdot\boldsymbol{u} = -dt\,D_t\rho/\rho$ is the fractional decrease in density (or increase in volume per unit mass) over a time dt. A strictly incompressible flow has the property that ρ is invariant for every parcel of fluid ($D_t\rho = 0$). Therefore, the velocity field in an incompressible flow satisfies $\boldsymbol{\nabla}\cdot\boldsymbol{u} = 0$.

2.2 Momentum Conservation

2.3 Boundary Conditions at a Fluid Interface

At every instant of time t, the two-dimensional interface is mathematically described by a functional constraint

$$F(\boldsymbol{r}, t) = 0\ . \tag{10}$$

In the case of small waves on the surface of a pond, we may write $F = z - h(x, y, t)$, where z is the vertical coordinate, and h is the wave height relative to the planar equilibrium surface attained in the absence of any disturbances. Similarly, the surface of a perturbed liquid drop can be represented by $r - r_s(\theta, \varphi, t)$, where r, θ, and φ are the usual spherical coordinates, and r_s is the distance to the surface along a particular direction. In both of these examples, F varies smoothly from negative values inside the liquid to positive values outside, a sign convention that we adopt here.

The gradient $\boldsymbol{\nabla}F$ evaluated at any position $\boldsymbol{r}$ is perpendicular to the level surface of F passing through $\boldsymbol{r}$. Over the special surface that defines the interface ($F = 0$), $\boldsymbol{\nabla}F$ is perpendicular to and points away from the liquid. Therefore,

$$\left(\frac{\boldsymbol{\nabla}F}{|\boldsymbol{\nabla}F|}\right)_{F=0} \equiv \hat{\boldsymbol{n}}_s \tag{11}$$

is the outward unit normal vector at the liquid surface.

We now explore the relationship between the kinematics of the fluid and the shape of the interface. As the fluid moves and the interface changes shape, the constraint $F = 0$ is preserved, by definition. Figure 1 depicts a portion of the moving surface at times t and $t + \delta t$. Also shown is the normal $\hat{\boldsymbol{n}}_s$ at some position $\boldsymbol{r}$ on the surface at time t. Let us draw a line through the point $\boldsymbol{r}$ parallel to $\hat{\boldsymbol{n}}_s$; this line pierces the displaced surface at position $\boldsymbol{r} + \zeta\hat{\boldsymbol{n}}_s$. After expanding $F(\boldsymbol{r} + \zeta\hat{\boldsymbol{n}}_s, t + \delta t)$ and using eq. (11), we find that, to lowest order in ζ and δt,

$$\zeta = -\delta t \frac{\partial_t F}{|\boldsymbol{\nabla}F|}\ , \tag{12}$$

where the derivatives are evaluated at $(\boldsymbol{r}, t)$.

2.4 Bernoulli Equation

2.5 Circulation

3 Small Oscillations of a Liquid Drop

- Static Equilibrium and Irrotational Flow

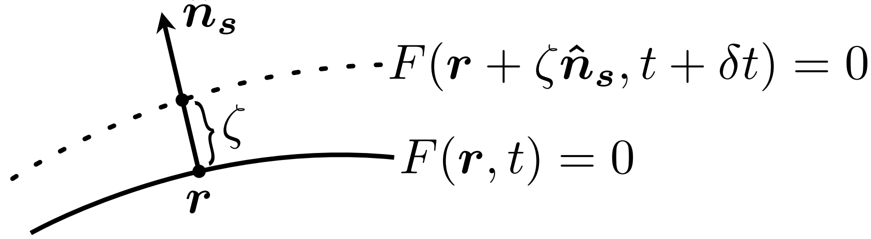

Figure 1: The fluid surface defined by $F = 0$ evolves with time. Along the line passing through $\boldsymbol{r}$ and parallel to $\hat{\boldsymbol{n}}_s$, the distance traversed by the surface in time interval δt is ζ.

- Linearized Equations (Bulk and Free Surface)
- Laplace's Equation in Spherical Coordinates
- Solution Constraints
- Normal-Mode Solution and Eigenfrequency
- Numerical Example of Capillary Frequency for Water

4 Total Mechanical Energy

- Kinetic Energy
- Surface Potential Energy (Deduce from constant KE, but then derive explicitly)
- The Sum (Digress on virial theorem, refer to Appendix)

Here we derive the total mechanical energy of a liquid drop undergoing small oscillations relative to an equilibrium spherical shape. Important results of the linear theory of inviscid, irrotational oscillation dynamics are presented. This is followed by complete calculations of both the total kinetic energy and the total potential energy stored in surface tension. Each of these quantities is inherently second order in small quantities, a fact that introduces a degree of subtlety, especially in the evaluation of the surface energy. We then sum the kinetic and surface energies to obtain the constant total mechanical energy.

4.1 Linear Oscillation Theory

Our discussion begins by considering the theory of small oscillations of an inviscid, incompressible, and irrotational liquid drop of density ρ and with surface tension σ. Relative to the drop's fixed center of mass, the radial distance to the free surface can be written as

$$r_s(\theta, \varphi, t) = a[1 + \eta(\theta, \varphi, t)] \ , \tag{13}$$

where a is a properly defined mean radius (to be investigated below), and $\eta \ll a$ is the dimensionless fractional displacement[2] along a radial ray with spherical angular coordinates (θ, φ). In a completely general fashion, we can expand η as a linear combination of spherical harmonics:

$$\eta(\theta, \varphi, t) = \sum_{\ell=0}^{\infty} \sum_{m=-\ell}^{\ell} b_\ell^m(t)\, Y_\ell^m(\theta, \varphi) \ , \tag{14}$$

[2]Hereafter we will use the notation $\mathcal{O}(\eta^n)$ in referring to quantities that are nth order in the mean amplitude of η or some other suitable dimensionless measure of smallness to which all dynamical variables are proportional.

where the Y_ℓ^m satisfy

$$Y_\ell^{m*} = (-1)^m Y_\ell^{-m} \tag{15}$$

and

$$\int_{4\pi} d\Omega\, Y_\ell^m Y_{\ell'}^{m'*} = 4\pi \delta_{\ell\ell'} \delta_{mm'} \;, \tag{16}$$

a normalization that proves convenient for our purposes. For real η, it is easily shown that

$$b_\ell^{m*} = (-1)^m b_\ell^{-m} \;. \tag{17}$$

We can immediately say something about the allowed range of ℓ without solving for the dynamics of the fluid. Because the fluid is incompressible, the drop's volume V_0 is a constant. Therefore, the drop can not undergo purely radial pulsations; thus, $b_0^m = 0$. Without loss of generality, we can also set $b_1^m = 0$, since an $\ell = 1$ mode corresponds merely to a shift of the origin, not a deformation of the drop relative to its center of mass. In other words, we adopt a coordinate system whose origin is fixed on the drop's center of mass. Hereafter we implicitly make the restriction $\ell \geq 2$ and write $\sum_{\ell,m}$ in place of $\sum_{\ell=2}^{\infty} \sum_{m=-\ell}^{\ell}$.

The drop's constant volume gives us a nontrivial constraint that relates a and η. The volume is given by the integral

$$V_0 = \int_{4\pi} d\Omega \int_0^{r_s} dr r^2 = \frac{a^3}{3} \int_{4\pi} d\Omega \left(1 + 3\eta + 3\eta^2 + \eta^3\right) \;. \tag{18}$$

Keeping up to $\mathcal{O}(\eta^2)$, we must evaluate the integrals $\int d\Omega\, \eta$ and $\int d\Omega\, \eta^2$. The $\mathcal{O}(\eta)$ integral is

$$\int_{4\pi} d\Omega\, Y_\ell^m \propto \int_{4\pi} d\Omega\, Y_\ell^m Y_0^0 \propto \delta_{\ell 0} \delta_{m0} \;, \tag{19}$$

which vanishes if $\ell \geq 2$. Since $\eta^2 = |\eta|^2$, we see that $\int d\Omega\, \eta^2$ is

$$\int_{4\pi} d\Omega\, |\eta|^2 = \sum_{\ell,m} \sum_{\ell',m'} b_\ell^m b_{\ell'}^{m'*} \int_{4\pi} d\Omega\, Y_\ell^m Y_{\ell'}^{m'*} = 4\pi \sum_{\ell,m} |b_\ell^m|^2 \;. \tag{20}$$

Thus, to second order, we find

$$V_0 = \frac{4\pi}{3} a^3 \left(1 + 3 \sum_{\ell,m} |b_\ell^m|^2\right) \;. \tag{21}$$

Now, let $V_0 = 4\pi a_0^3/3$, where a_0 is interpreted as the radius of the drop at rest. It follows that, to second order,

$$a = a_0 \left(1 - \sum_{\ell,m} |b_\ell^m|^2\right) \;. \tag{22}$$

While this second-order discrepancy between a and a_0 may seem unimportant in a linear theory of drop oscillations, we will see below that eq. (22) is needed in order to correctly calculate the surface area of the deformed drop.

We now return to the core problem of the oscillatory dynamics of the drop. The assumptions of incompressibility and irrotational dynamics imply that the fluid velocity field $\boldsymbol{u}$ satisfies

$$\boldsymbol{\nabla} \cdot \boldsymbol{u} = \boldsymbol{\nabla} \times \boldsymbol{u} = 0 \;. \tag{23}$$

The vanishing of the curl means that we can write $\boldsymbol{u} = \boldsymbol{\nabla}\psi$, where $\psi(r, \varphi, t)$ is the scalar velocity potential. Because $\boldsymbol{\nabla} \cdot \boldsymbol{u} = 0$, we then have

$$\nabla^2 \psi = 0 \ , \tag{24}$$

which is the fundamental equation to be solved, subject to the appropriate boundary conditions. Seeking a normal-mode solution of the form

$$\psi = e^{j\omega t} f(r, \theta, \varphi) \ , \tag{25}$$

we see that f is a solution of Laplace's equation ($\nabla^2 f = 0$) in spherical coordinates. The non-singular spatial eigenfunction is

$$f_\ell^m(r, \theta, \varphi) \propto r^\ell Y_\ell^m(\theta, \varphi) \ . \tag{26}$$

The dynamic boundary conditions at the free surface of the drop, which we do not derive here (see, e.g., Landau & Lifshitz 1987, pg. 246), provide the eigenfrequency,

$$\omega_\ell^2 = \ell(\ell - 1)(\ell + 2)\omega_c^2 \ , \tag{27}$$

where

$$\omega_c = \left(\frac{\sigma}{\rho a^3} \right)^{1/2} \ , \tag{28}$$

known as the capillary frequency, contains the fundamental parameter scalings. Note that the eigenfrequency does not depend on the azimuthal number m. We see that ω_ℓ vanishes for $\ell = 0$ and $\ell = 1$, as it must according to the remarks following eq. (17). So, the lowest oscillation frequency is attained when $\ell = 2$, a quadrupolar distortion, such that $\omega_2 = \sqrt{8}\omega_c$.

We can now write the full solution for the velocity potential:

$$\psi(r, \theta, \varphi) = a \sum_{\ell, m} c_\ell^m(t) \left(\frac{r}{a} \right)^\ell Y_\ell^m(\theta, \varphi) \ . \tag{29}$$

Just as for the b_ℓ^m in the expansion of η, a real ψ implies

$$c_\ell^{m*} = (-1)^m c_\ell^{-m} \ . \tag{30}$$

Each ℓ corresponds to a normal mode with frequency ω_ℓ, so that

$$c_\ell^m(t) = \alpha_\ell^m e^{j\omega_\ell t} + \beta_\ell^m e^{-j\omega_\ell t} \ , \tag{31}$$

where α_ℓ^m and β_ℓ^m are complex constants with units of velocity, and

$$\alpha_\ell^{m*} = (-1)^m \beta_\ell^{-m} \ ; \quad \beta_\ell^{m*} = (-1)^m \alpha_\ell^{-m} \tag{32}$$

according to eq. (30).

It is clear that the b_ℓ^m and c_ℓ^m are mathematically related. To make this explicit, we cite the kinematic surface boundary condition:

$$u_r|_{r=r_s} = \partial_r \psi|_{r=r_s} = a\partial_t \eta + \mathcal{O}(\eta^2) \ , \tag{33}$$

which yields the simple differential equation

$$a\partial_t b_\ell^m = \ell c_\ell^m \ , \tag{34}$$

with the solution

$$b_\ell^m(t) = \frac{-j\ell}{a\omega_\ell} \left(\alpha_\ell^m e^{j\omega_\ell t} - \beta_\ell^m e^{-j\omega_\ell t} \right) \ . \tag{35}$$

It will be useful later to have a relation between $|b_\ell^m|^2$ and $|c_\ell^m|^2$; this is

$$|b_\ell^m|^2 = \left(\frac{\ell}{a\omega_\ell}\right)^2 \left[|c_\ell^m|^2 - 2k_\ell^m\right] \ , \tag{36}$$

where

$$k_\ell^m = \alpha_\ell^m \beta_\ell^{m*} e^{2j\omega_\ell t} + \alpha_\ell^{m*} \beta_\ell^m e^{-2j\omega_\ell t} \ , \tag{37}$$

and

$$|c_\ell^m|^2 = |\alpha_\ell^m|^2 + |\beta_\ell^m|^2 + k_\ell^m \ . \tag{38}$$

4.2 Kinetic Energy

From ψ we can derive every dynamical quantity of interest. We are principally concerned here with the energy of the drop. To this end, we first calculate the total kinetic energy, given by

$$E_{\mathrm{K}}(t) = \int\limits_{V_d(t)} dV \, \tfrac{1}{2}\rho u^2 \ , \tag{39}$$

where $V_d(t)$ denotes the variable 3-dimensional domain occupied by the fluid; the surface boundary of the drop is denoted by $A_d(t)$. Note that

$$u^2 = \boldsymbol{\nabla}\psi \cdot \boldsymbol{\nabla}\psi = \boldsymbol{\nabla} \cdot (\psi \boldsymbol{\nabla}\psi) - \psi \nabla^2 \psi \ . \tag{40}$$

We know that $\nabla^2\psi = 0$, and thus

$$E_{\mathrm{K}}(t) = \tfrac{1}{2}\rho \int\limits_{V_d(t)} dV \, \boldsymbol{\nabla} \cdot (\psi \boldsymbol{\nabla}\psi) \ . \tag{41}$$

By the divergence theorem, this becomes an integral over the drop's surface,

$$E_{\mathrm{K}}(t) = \tfrac{1}{2}\rho \int\limits_{A_d(t)} dA \, \hat{\boldsymbol{n}} \cdot (\psi \boldsymbol{\nabla}\psi) \ , \tag{42}$$

where dA is an element of surface area, $\hat{\boldsymbol{n}}$ is the outward surface normal, and the integrand is evaluated at $r = a$. Since $\psi\boldsymbol{\nabla}\psi$ in eq. (42) is already $\mathcal{O}(\eta^2)$, we can let $dS = d\Omega\, a^2$ and $\hat{\boldsymbol{n}} = \hat{\boldsymbol{r}}$, so that

$$E_{\mathrm{K}}(t) = \tfrac{1}{2}\rho a^2 \int\limits_{4\pi} d\Omega \, \psi \partial_r \psi \ , \tag{43}$$

Substituting for ψ from eq. (29) and using eq. (30), we obtain

$$E_{\mathrm{K}}(t) = 2\pi\rho a^3 \sum_{\ell,m} \ell |c_\ell^m|^2 \ . \tag{44}$$

We can freely replace a with a_0 above, because $|c_\ell^m|^2$ is $\mathcal{O}(\eta^2)$.

4.3 Surface Energy

When at rest, the drop adopts a spherical shape. Deviations from this equilibrium shape have a larger total area A (at constant volume) and surface potential energy, given by

$$E_{\mathrm{S}}(t) = \sigma A(t) \ . \tag{45}$$

As long as the surface remains convex, the area is exactly

$$A = \int_{4\pi} \frac{d\Omega r_s^2}{\hat{\boldsymbol{n}} \cdot \hat{\boldsymbol{r}}} \ , \tag{46}$$

where $d\Omega$ is an element of solid angle, and $\hat{\boldsymbol{r}}$ and $\hat{\boldsymbol{n}}$ are the unit radial vector and unit surface normal, respectively. Before we can proceed, we need the full expression for $\hat{\boldsymbol{n}}$ in terms of η. The surface is defined through the functional constraint

$$F(r, \theta, \varphi, t) = r - a[1 + \eta(\theta, \varphi, t)] = 0 \ . \tag{47}$$

The unit surface normal is $\hat{\boldsymbol{n}} = (\boldsymbol{\nabla} F / |\boldsymbol{\nabla} F|)_{r=r_s}$, where

$$\boldsymbol{\nabla} F|_{r=r_s} = \hat{\boldsymbol{r}} - \frac{1}{(1+\eta)} \left(\hat{\boldsymbol{\theta}} \partial_\theta \eta + \frac{\hat{\boldsymbol{\varphi}}}{\sin\theta} \partial_\varphi \eta \right) \ . \tag{48}$$

Therefore, $\hat{\boldsymbol{n}} \cdot \hat{\boldsymbol{r}} = |\boldsymbol{\nabla} F|^{-1}_{r=r_s}$, and, to $\mathcal{O}(\eta^2)$, the area is

$$A = a^2 \int_{4\pi} d\Omega \left\{ 1 + 2\eta + \eta^2 + \tfrac{1}{2} \left[(\partial_\theta \eta)^2 + \frac{1}{\sin^2\theta} (\partial_\varphi \eta)^2 \right] \right\} \tag{49}$$

We have already shown that $\int_{4\pi} d\Omega\, \eta = 0$ (see eq. [19]), and so deviations of A from $4\pi a^2$ are inherently $\mathcal{O}(\eta^2)$. We have also already evaluated $\int_{4\pi} d\Omega\, \eta^2$ in eq. (20). The final piece to be determined is the integral of the term in square brackets in eq. (49), which is $a^2(\boldsymbol{\nabla}\eta \cdot \boldsymbol{\nabla}\eta)_{r=r_s}$ with $\partial_r \eta = 0$, where (see eq. [40])

$$\boldsymbol{\nabla}\eta \cdot \boldsymbol{\nabla}\eta = \boldsymbol{\nabla} \cdot (\eta \boldsymbol{\nabla}\eta) - \eta \nabla^2 \eta \ . \tag{50}$$

Let $\boldsymbol{G} = \eta \boldsymbol{\nabla} \eta$, so that integration of the above divergence term gives

$$\int_{4\pi} d\Omega \left[\frac{1}{\sin\theta} \partial_\theta (\sin\theta G_\theta) + \frac{1}{\sin\theta} \partial_\varphi G_\varphi \right] \quad = \quad \int_0^{2\pi} d\varphi \, \left[\sin\theta G_\theta \right]_{\theta=0}^{\theta=\pi} \ + \ \int_0^{\pi} d\theta \, \left[G_\varphi \right]_{\varphi=0}^{\varphi=2\pi} \quad = \quad 0 \quad . \tag{51}$$

The integral vanishes because the surface is closed. We are now left with

$$\int_{4\pi} d\Omega \left[(\partial_\theta \eta)^2 + \frac{1}{\sin^2\theta} (\partial_\varphi \eta)^2 \right] = - \int_{4\pi} d\Omega\, \eta \, \left[\frac{1}{\sin\theta} \partial_\theta (\sin\theta \partial_\theta \eta) + \frac{1}{\sin^2\theta} \partial_\varphi^2 \eta \right] \ . \tag{52}$$

Denote the bracketed term on the right-hand side by $L^2\eta$, where L^2 is the angular Laplace operator. Substituting the spherical harmonic expansion of η, we find

$$- \int_{4\pi} d\Omega\, \eta\, L^2 \eta = - \sum_{\ell,m} b_\ell^m \int_{4\pi} d\Omega\, \eta\, (L^2 Y_\ell^m) \ . \tag{53}$$

Spherical harmonics are defined through the differential equation

$$L^2 Y_\ell^m + \ell(\ell+1) Y_\ell^m = 0 \ , \tag{54}$$

and thus

$$\begin{aligned} - \int_{4\pi} d\Omega\, \eta\, L^2 \eta \quad &= \quad \sum_{\ell,m} b_\ell^m \ell(\ell+1) \int_{4\pi} d\Omega\, \eta\, Y_\ell^m \\ &= \quad \sum_{\ell,m} b_\ell^m \ell(\ell+1) \sum_{\ell',m'} b_{\ell'}^{m'} \int_{4\pi} d\Omega Y_{\ell'}^{m'} Y_\ell^m \\ &= \quad 4\pi \sum_{\ell,m} \ell(\ell+1) |b_\ell^m|^2 \ . \end{aligned} \tag{55}$$

The area is now

$$A = 4\pi a^2 \left\{ 1 + \sum_{\ell,m} |b_\ell^m|^2 \left[1 + \tfrac{1}{2}\ell(\ell+1)\right] \right\} . \tag{56}$$

Because we are retaining second-order terms in the evaluation of A, consistency demands that we incorporate the expression for a in eq. (22); this gives

$$A = 4\pi a_0^2 \left[1 + \tfrac{1}{2} \sum_{\ell,m} |b_\ell^m|^2 (\ell-1)(\ell+2) \; , \right] . \tag{57}$$

which shows correctly that for $\ell = 1$ the area takes its rest value. Dropping the constant term in A and replacing $|b_\ell^m|^2$ with the expression in eq. (36), we obtain

$$E_{\rm S}(t) = 2\pi \sum_{\ell,m} \frac{\sigma \ell^2}{\omega_\ell^2} (\ell-1)(\ell+2) \left[|c_\ell^m|^2 - 2k_\ell^m \right] . \tag{58}$$

We can now proceed to compute the total mechanical energy of the drop.

4.4 The Sum

After substitution of eqs. (27) and (28) for the eigenfrequency, the surface energy becomes

$$E_{\rm S}(t) = 2\pi\rho a_0^3 \sum_{\ell,m} \ell \left[|c_\ell^m|^2 - 2k_\ell^m \right] . \tag{59}$$

Adding this to the kinetic energy and making use of eq. (38), we derive the constant total energy,

$$\begin{aligned} E_{\rm M} = E_{\rm K}(t) + E_{\rm S}(t) &= 4\pi\rho a_0^3 \sum_{\ell,m} \ell \left[|c_\ell^m|^2 - k_\ell^m \right] \\ &= 4\pi\rho a_0^3 \sum_{\ell,m} \ell \left[|\alpha_\ell^m|^2 + |\beta_\ell^m|^2 \right] . \end{aligned} \tag{60}$$

The same result is obtained if we compute the time average of $2E_{\rm K}$:

$$\langle 2E_{\rm K} \rangle \equiv \lim_{T\to\infty} \frac{1}{T} \int_0^T dt\, 2E_{\rm K}(t) = 4\pi\rho a_0^3 \sum_{\ell,m} \ell \left[|\alpha_\ell^m|^2 + |\beta_\ell^m|^2 + \lim_{T\to\infty} \frac{1}{T} \int_0^T dt\, k_\ell^m \; . \right] \tag{61}$$

The term k_ℓ^m is a sum of pure sinusoids and so its long-time average vanishes. The simple result $E_{\rm M} = \langle 2E_{\rm K} \rangle$ is essentially a manifestation of the virial theorem for a mechanical system whose constituents are bound by conservative forces that vary linearly in the spatial coordinates, as in the case of linear oscillations about a stable equilibrium configuration.

5 Viscous Dissipation

Up to this point, we have considered incompressible, inviscid, irrotational flow. Under these conditions, we can analytically solve for the velocity potential in the linear theory of an oscillating liquid drop. The corresponding total mechanical energy is conserved. When viscosity is introduced, the mechanical energy decreases with time as friction converts kinetic energy of the fluid into heat.

If the viscosity is sufficiently small, damping of the energy is significant only after a time much longer than ω_c^{-1} (eq. [28]), the fundamental dynamical time scale of the drop. Over short times, neither the total energy nor the flow field show appreciable deviations from the inviscid results. In what follows, we derive a general expression for the rate at which mechanical energy is lost from a viscous liquid drop, first for a compressible fluid and then specializing to the incompressible case. We then substitute in this expression the inviscid, irrotational solution described above. This prescription effectively assumes that the viscous dissipation is merely a spectator during the inviscid oscillations, which is equivalent to the assumption of very small viscosity. Although heat is generated at the expense of mechanical energy, we do not consider any thermal effects.

5.1 Dissipation Integral

In flux conservative form, the equation for fluid momentum conservation is

$$\partial_t(\rho u_i) + \partial_j(\rho u_i u_j) = \partial_j T_{ij} \ , \tag{62}$$

where T_{ij} is the (symmetric) stress tensor, and we have neglected body forces such as gravity. Note that ρu_i is the x_i component of the momentum density, and $\rho u_i u_j - T_{ij}$ is the net flux of x_i momentum in the x_j direction. The stress tensor is a sum of the isotropic pressure p and viscous stress:

$$T_{ij} = -p\delta_{ij} + \tau_{ij} \ . \tag{63}$$

Using the equation for mass conservation, $\partial_t\rho + \boldsymbol{\nabla}\cdot(\rho\boldsymbol{u}) = 0$, we transform eq. (62) into the general form of the Navier-Stokes equation:

$$\rho(\partial_t u_i + u_j\partial_j u_i) = -\partial_i p + \partial_j\tau_{ij} \ . \tag{64}$$

Multiplying by u_i and carrying out some manipulations, we obtain

$$\partial_t\left(\tfrac{1}{2}\rho u^2\right) + \partial_j\left(u_j\,\tfrac{1}{2}\rho u^2\right) = -u_i\partial_i p + u_i\partial_j\tau_{ij} \ , \tag{65}$$

an equation for the continuity of kinetic energy in flux conservative form. With the substitutions

$$u_i\partial_i p = \partial_i(u_i p) - p\partial_i u_i \quad ; \quad u_i\partial_j\tau_{ij} = \partial_j(u_i\tau_{ij}) - \tau_{ij}\partial_j u_i \ , \tag{66}$$

volume integration and the divergence theorem yield

$$\begin{aligned}
\int\limits_{V_d(t)} dV\,\partial_t\left(\tfrac{1}{2}\rho u^2\right) &= -\int\limits_{V_d(t)} dV\,\partial_j\left[u_j\left(\tfrac{1}{2}\rho u^2 + p) - u_i\tau_{ij}\right)\right] + \int\limits_{V_d(t)} dV[-\tau_{ij}\partial_j u_i + p\partial_j u_j] \\
&= -\int\limits_{A_d(t)} dA\,n_i u_i\left(\tfrac{1}{2}\rho u^2\right) - \int\limits_{A_d(t)} dA\,n_j(u_j p - u_i\tau_{ij}) \\
&\quad + \int\limits_{V_d(t)} dV[-\tau_{ij}\partial_j u_i + p\partial_j u_j] \ ,
\end{aligned} \tag{67}$$

where n_i is the x_i component of $\hat{\boldsymbol{n}}$. Recalling that $E_{\rm K} = \int_{V_d(t)} \frac{1}{2}\rho u^2$ and using eq. (A18) from the Appendix, we find

$$\frac{dE_{\rm K}}{dt} = \int\limits_{V_d(t)} dV\,\partial_t\left(\tfrac{1}{2}\rho u^2\right) + \int\limits_{A_d(t)} dA\,n_i u_i\left(\tfrac{1}{2}\rho u^2\right) \ . \tag{68}$$

Equation (67) now reads

$$\frac{dE_{\rm K}}{dt} = -\int\limits_{A_d(t)} dA\,n_j(u_j p - u_i\tau_{ij}) + \int\limits_{V_d(t)} dV[-\tau_{ij}\partial_j u_i + p\partial_j u_j] \ . \tag{69}$$

For an inviscid, incompressible fluid, the volume integral above vanishes, and we must recover the statement of total mechanical energy conservation, namely, $dE_{\rm K}/dt = -dE_{\rm S}/dt$. In order to see that this does indeed hold, we must discuss the surface boundary conditions on the stress.

By Newton's third law, the force transmitted from fluid 1 to fluid 2 across the interface is equal in magnitude and opposite in direction to the force exerted by fluid 2 on fluid 1. For an interfacial element of fluid 1, the x_i component of the force per unit area transmitted in the direction of the local normal $\hat{\boldsymbol{n}}^{(1)}$ (from 1 to 2) is

$$P_i^{(1)} = -T_{ij}^{(1)} n_j^{(1)} - \sigma \left(\partial_j n_j^{(1)} \right) n_i^{(1)} \ . \tag{70}$$

Here the minus sign on the surface-tension term accounts for the fact that the corresponding force points in to the fluid when $\boldsymbol{\nabla} \cdot \hat{\boldsymbol{n}} > 0$ (i.e., when the interface bulges outward relative to the fluid). This element feels a force $-T_{ij}^{(2)} n_j^{(2)}$ from an adjacent element of fluid 2, where $\hat{\boldsymbol{n}}^{(2)} = -\hat{\boldsymbol{n}}^{(1)}$ is the normal from 2 to 1. From the force balance condition, we have

$$-\left[T_{ij}^{(1)} n_j^{(1)} + T_{ij}^{(2)} n_j^{(2)} \right] - \sigma \left(\partial_j n_j^{(1)} \right) n_i^{(1)} = 0 \ . \tag{71}$$

Using the definition of the stress tensor (eq. [63]) and dropping the superscripts, we obtain

$$\left[p^{(1)} - p^{(2)} - \sigma \left(\partial_j n_j^{(1)} \right) \right] n_i^{(1)} = \left[\tau_{ij}^{(1)} - \tau_{ij}^{(2)} \right] n_j^{(1)} \ . \tag{72}$$

In the absence of viscous stress, we recover the Young-Laplace relation for the pressure jump due to surface tension. If fluid 2 is replaced by a vacuum, and we drop the superscripts, the surface stress condition reads

$$p n_i - \tau_{ij} n_j = \sigma \left(\partial_j n_j \right) n_i \ . \tag{73}$$

Inspection of the area integral in eq. (69) and use of eq. (73) reveals that

$$n_j (u_j p - u_i \tau_{ij}) = u_i (p n_i - \tau_{ij} n_j) = \sigma \, \hat{\boldsymbol{n}} \cdot \boldsymbol{u} \, \boldsymbol{\nabla} \cdot \hat{\boldsymbol{n}} \ , \tag{74}$$

and thus the rate of change of the drop's kinetic energy is

$$\frac{dE_{\rm K}}{dt} = -\sigma \int\limits_{A_d(t)} dA \, \hat{\boldsymbol{n}} \cdot \boldsymbol{u} \, \boldsymbol{\nabla} \cdot \hat{\boldsymbol{n}} + \int\limits_{V_d(t)} dV [-\tau_{ij} \partial_j u_i + p \boldsymbol{\nabla} \cdot \boldsymbol{u}] \ . \tag{75}$$

With the help of eq. (A20) in the Appendix, we can derive an exact expression for the rate of change of drop area:

$$\frac{dA}{dt} = \frac{d}{dt} \int\limits_{A_d(t)} dA \, \hat{\boldsymbol{n}} \cdot \hat{\boldsymbol{n}} = \int\limits_{A_d(t)} dA \, \hat{\boldsymbol{n}} \cdot \partial_t \hat{\boldsymbol{n}} + \int\limits_{A_d(t)} dA \, \hat{\boldsymbol{n}} \cdot \boldsymbol{u} \, \boldsymbol{\nabla} \cdot \hat{\boldsymbol{n}} \ . \tag{76}$$

Noting that

$$\hat{\boldsymbol{n}} \cdot \partial_t \hat{\boldsymbol{n}} = \tfrac{1}{2} \partial_t (\hat{\boldsymbol{n}} \cdot \hat{\boldsymbol{n}}) = 0 \ , \tag{77}$$

we now identify the area integral in eq. (75) as $-dE_{\rm S}/dt$, which allows us to write

$$\frac{dE_{\rm M}}{dt} = \int\limits_{V_d(t)} dV [-\tau_{ij} \partial_j u_i + p \boldsymbol{\nabla} \cdot \boldsymbol{u}] \tag{78}$$

for the rate of change of mechanical energy.

In an incompressible fluid,

$$\frac{dE_{\rm M}}{dt} = -\int\limits_{V_d(t)} dV \tau_{ij} \partial_j u_i \ . \tag{79}$$

From the symmetry of τ_{ij}, we have

$$\tau_{ij} \partial_j u_i = \tau_{ij} \frac{1}{2} \left(\partial_i u_j + \partial_j u_i \right) \equiv \tau_{ij} e_{ij} \ , \tag{80}$$

where e_{ij} is the symmetric strain rate tensor. For a Newtonian fluid[3], it can be shown (see Landau & Lifshitz, pg. 45; Kundu & Cohen, pg. 95) that when $\boldsymbol{\nabla}\cdot\boldsymbol{u}=0$,

$$\tau_{ij} = 2\mu e_{ij} \ , \tag{81}$$

so that

$$\frac{dE_{\rm M}}{dt} = -2\mu \int\limits_{V_d(t)} dV e_{ij}e_{ij} \ . \tag{82}$$

5.2 Spectator Limit

Our present task is to substitute the inviscid, irrotational solution for the oscillating drop (eq. [29]) into the energy dissipation rate given by eq. (82). Before we do this, we massage eq. (82) into a more convenient form. As already discussed, $\boldsymbol{u}=\boldsymbol{\nabla}\psi$ and $\nabla^2\psi=0$ in an incompressible, irrotational flow. It follows that

$$\partial_j u_i = \partial_j\partial_i\psi = \partial_i\partial_j\psi = \partial_i u_j \ , \tag{83}$$

and so,

$$e_{ij}e_{ij} = \partial_i u_j \partial_i u_j \ . \tag{84}$$

Substituting this into eq. (82) and integrating by parts, we find

$$\frac{dE_{\rm M}}{dt} = -2\mu \int\limits_{V_d(t)} dV \left[\boldsymbol{\nabla}\cdot\left(\boldsymbol{\nabla}\tfrac{1}{2}u^2\right) - \boldsymbol{u}\cdot\left(\nabla^2\boldsymbol{u}\right)\right] \ . \tag{85}$$

The second term in the above integrand vanishes because $\nabla^2 u_i = \partial_i\nabla^2\psi = 0$, while the divergence theorem applied to the first term yields

$$\frac{dE_{\rm M}}{dt} = -\mu \int\limits_{A_d(t)} dA\, \hat{\boldsymbol{n}}\cdot\boldsymbol{\nabla}u^2 \ . \tag{86}$$

Not only is this expression simpler than eq. (82), it is now index-free and manifestly independent of coordinate system.

At $\mathcal{O}(\eta^2)$, eq. (86) becomes

$$\frac{dE_{\rm M}}{dt} = -\mu a_0^2\, \partial_r \int\limits_{4\pi} d\Omega\, u^2 \ . \tag{87}$$

Recall that $u^2 = \boldsymbol{\nabla}\cdot(\psi\boldsymbol{\nabla}\psi)$ (see eq. [40]). Terms in the divergence with angular derivatives vanish under the integral (see eq. [51]), which leaves

$$\frac{dE_{\rm M}}{dt} = -\mu a_0^2\, \partial_r r^{-2}\partial_r r^2 \int\limits_{4\pi} d\Omega\, \psi\partial_r\psi \ . \tag{88}$$

It is understood that this expression is ultimately evaluated at $r=r_s$. The material in § 4.2 (eqs. [43] and [44]) shows that

$$\int\limits_{4\pi} d\Omega\, \psi\partial_r\psi = 4\pi a_0 \sum_{\ell,m} \left(\frac{r}{a_0}\right)^{2\ell-1} \ell|c_\ell^m|^2 \ , \tag{89}$$

[3]In a Newtonian fluid, there is a strictly linear relationship between the viscous stress and the rate of strain; i.e., we have $\tau_{ij} = c_{ijkl}e_{kl}$, where the c_{ijkl} are constant coefficients.

and thus,

$$\frac{dE_{\rm M}}{dt} = -8\pi a_0 \mu \sum_{\ell,m} \ell(\ell-1)(2\ell+1)|c_\ell^m|^2 \ . \tag{90}$$

A more practical expression for the energy dissipation rate is obtained by averaging over a time $T \gg \omega_c^{-1}$:

$$\left\langle \frac{dE_{\rm M}}{dt} \right\rangle_T = -\, 8\pi a_0 \mu \sum_{\ell,m} \ell(\ell-1)(2\ell+1) \left[\frac{1}{T} \int_0^T dt |c_\ell^m|^2 \right] \ . \tag{91}$$

To a good approximation, we can replace the term in square brackets with its asymptotic limit, $|\alpha_\ell^m|^2 + |\beta_\ell^m|^2$ (see eq. [61]). The time at which the mechanical energy lost equals the initial value is

$$T_{\rm dis} = \frac{\rho a_0^2}{2\mu} \frac{\sum_{\ell,m} \ell \left[|\alpha_\ell^m|^2 + |\beta_\ell^m|^2 \right]}{\sum_{\ell,m} \ell(\ell-1)(2\ell+1) \left[|\alpha_\ell^m|^2 + |\beta_\ell^m|^2 \right]} \ , \tag{92}$$

which contains the scaling $T_{\rm dis} \propto \rho a_0^2/\mu$ expected on dimensional grounds. Compare this damping time to the results in Lamb (1945, pg. 640).

A useful dimensionless dissipation time is

$$T_{\rm dis}\omega_c \propto \frac{(\sigma \rho a_0)^{1/2}}{\mu} \ , \tag{93}$$

which is essentially the number of dynamical times before a significant fraction of the initial mechanical energy is transformed into heat. The dimensionless Ohnesorge number (Oh) is defined as $\mu/(\sigma\rho a_0)^{1/2}$, the inverse of the quantity above. Although Oh is often encountered in discussions of the dynamics of viscous drops, we have not come across the clear physical interpretation of Oh given here. Specifically, Oh^{-1} is of order the number of capillary oscillation periods over which the mechanical energy decreases by a significant factor. A large value of Oh^{-1} indicates small viscosity. This is analogous to the quality factor Q for a damped harmonic oscillator, which is proportional to the time scale for the oscillator to "ring down."

A Derivatives of Integrals with Variable Domains

Here we develop formulae for the time derivatives of integrals over a variable volume and its bounding surface. Although the analysis below is complete and fairly rigorous, it is also somewhat abstract. It was realized after this material was written that there is a more elementary technique that arrives at the same conclusions. This alternative approach is not presented here. The interested reader should examine the proof of Kelvin's Circulation Theorem in Kundu & Cohen (2002, pg. 130), where one takes the material time derivative of a closed line integral. The same mathematical method used there can be readily generalized to a three-dimensional volume.

A.1 Elementary Derivation

The integrals of interest have the form

$$I_{V_d}(t) = \int\limits_{V_d(t)} dV g(\boldsymbol{r}, t) \ , \tag{A1}$$

and

$$I_{A_d}(t) = \int\limits_{A_d(t)} d\boldsymbol{A} \cdot \boldsymbol{G}(\boldsymbol{r}_s, t) \ , \tag{A2}$$

where $V_d(t)$ represents the time-dependent volumetric domain, $A_d(t)$ is the bounding surface of $V_d(t)$, $d\boldsymbol{A} = dA\hat{\boldsymbol{n}}$ is the vector surface area element, and $\boldsymbol{r}$ and $\boldsymbol{r}_s$ represent coordinate vectors within and on the surface of the drop, respectively. We assume that the functions g and $\boldsymbol{G}$ are continuous and differentiable throughout V_d.

A.2 Delta Function Technique

In order to better understand the nature of the problem at hand, we consider the one-dimensional version of the above integrals:

$$I_{[x_1,x_2]}(t) = \int_{x_1(t)}^{x_2(t)} dx\, g(x,t) \ . \tag{A3}$$

Application of the Leibniz integral rule yields the time derivative of $I_{[x_1,x_2]}$:

$$\frac{d}{dt} I_{[x_1,x_2]} = \int_{x_1(t)}^{x_2(t)} dx\, \partial_t g(x,t) + g(x_2(t),t)\, \partial_t x_2 - g(x_1(t),t)\, \partial_t x_1 \ . \tag{A4}$$

The result includes an integral over the time derivative of the function as well as terms containing the rates of change of the boundary points.

Before we attempt to generalize eq. (A4) to higher dimensions, we show an alternative representation of $I_{[x_1,x_2]}$. It is possible to shift the burden of variable boundary conditions from the limits of the integral to the integrand. Define a weighting function W with the properties

$$W(x;x_1,x_2) = \begin{cases} 1 \ , & x_1 \le x \le x_2 \\ 0 \ , & \text{else} \end{cases} \ , \tag{A5}$$

so that

$$I_{[x_1,x_2]}(t) = \int_{-\infty}^{\infty} dx\, W(x;x_1,x_2) f(x,t) \ . \tag{A6}$$

Now the time derivative is simply

$$\frac{d}{dt} I_{[x_1,x_2]} = \int_{-\infty}^{\infty} dx\, \partial_t [W(x;x_1,x_2) g(x,t)] \ . \tag{A7}$$

We can write W as

$$W(x;x_1,x_2) = H(x-x_1) - H(x-x_2) \ , \tag{A8}$$

where H is the Heaviside step function defined by

$$H(x) = \int_{-\infty}^{x} dz\, \delta(z) = \begin{cases} 0 \ , & x < 0 \\ \frac{1}{2} \ , & x = 0 \\ 1 \ , & x > 0 \end{cases} \ . \tag{A9}$$

The fact that $H(\frac{1}{2})$ is $\frac{1}{2}$ rather than 1 is immaterial in the present discussion. The time derivative of W is

$$\partial_t W(x;x_1,x_2) = -\delta(x-x_1)\, \partial_t x_1 + \delta(x-x_2)\, \partial_t x_2 \ . \tag{A10}$$

Substituting eq. (A8) into eq. (A7) and using eq. (A10), we obtain the correct result in eq. (A4).

To apply these ideas to volume and surface integrals, we first need a functional representation of the closed surface. This is given by the constraint

$$F(\boldsymbol{r},t) = 0 \ , \tag{A11}$$

or $F = r - r_s = 0$ in spherical coordinates, as in eq. (47), where $r_s(\theta, \varphi, t)$ is the surface radius. Just as in the spherical representation, let us define $F(\boldsymbol{r}, t)$ such that $F > 0$ outside V_d and $F \leq 0$ inside (equality at the surface). The general volume integral can now be recast as

$$I_{V_d}(t) = \int\limits_{V_d(t)} dV g = \int\limits_{\infty} dV H(-F)\, g \ , \tag{A12}$$

where $H(-F)$ is the weighting function analogous to eq. (A8) that restricts the integrand to nonzero values only inside V_d, and $\int_\infty dV$ denotes a volume integral over all space. The time derivative of the transformed volume integral is

$$\frac{d}{dt} I_{V_d}(t) = \int\limits_{V_d(t)} dV \partial_t g - \int\limits_{\infty} dV \delta(F) \partial_t F\, g \ . \tag{A13}$$

If we follow a point $\boldsymbol{r}_s(t)$ that comoves with the variable surface, the total derivative of F satisfies

$$\frac{dF}{dt} = \partial_t F + \frac{d\boldsymbol{r}_s}{dt} \cdot \boldsymbol{\nabla} F = 0 \ , \tag{A14}$$

and thus the second term on the right-hand side of eq. (A13) becomes

$$-\int\limits_{\infty} dV \delta(-F) \partial_t F\, g = \int\limits_{\infty} dV \delta(F) \left(\frac{d\boldsymbol{r}_s}{dt} \cdot \boldsymbol{\nabla} F \right) g \ . \tag{A15}$$

The delta function above essentially restricts the integration to the surface, a notion that we can turn into a rigorous statement as follows.

Using the divergence theorem and eq. (A12), we rewrite the integral over the closed[4] boundary of V_d as

$$I_{A_d}(t) = \int\limits_{A_d(t)} d\boldsymbol{A} \cdot \boldsymbol{G} = \int\limits_{V_d(t)} dV \boldsymbol{\nabla} \cdot \boldsymbol{G} = \int\limits_{\infty} dV H(-F) \boldsymbol{\nabla} \cdot \boldsymbol{G} \ . \tag{A16}$$

Integration by parts gives

$$\int\limits_{\infty} dV H(-F) \boldsymbol{\nabla} \cdot \boldsymbol{G} = \int\limits_{\infty} dV \boldsymbol{\nabla} \cdot [H(-F) \boldsymbol{G}] + \int\limits_{\infty} dV \delta(F) \boldsymbol{\nabla} F \cdot \boldsymbol{G} \tag{A17}$$

The volume integral of $\boldsymbol{\nabla} \cdot [H(-F)\boldsymbol{G}]$ becomes a surface integral at infinity, which vanishes because $H = 0$ outside the drop. Therefore, we see that

$$\int\limits_{A_d(t)} d\boldsymbol{A} \cdot \boldsymbol{G} = \int\limits_{\infty} dV \delta(F) \boldsymbol{\nabla} F \cdot \boldsymbol{G} \tag{A18}$$

for arbitrary $\boldsymbol{G}$. Applying this relationship to eq. (A15), we find

$$\frac{d}{dt} I_{V_d}(t) = \int\limits_{V_d(t)} dV \partial_t g + \int\limits_{A_d(t)} d\boldsymbol{A} \cdot \frac{d\boldsymbol{r}_s}{dt}\, g \ , \tag{A19}$$

which makes explicit the connection to the one-dimensional result in eq. (A4). In the subject of fluid dynamics, this result is referred to as Reynolds' transport theorem.

Starting with the second equality in eq. (A16), we can obtain dI_{A_d}/dt directly from eq. (A19) by replacing g with $\boldsymbol{\nabla} \cdot \boldsymbol{G}$ and using the divergence theorem:

$$\frac{d}{dt} I_{A_d}(t) = \int\limits_{A_d(t)} d\boldsymbol{A} \cdot (\partial_t \boldsymbol{G}) + \int\limits_{A_d(t)} d\boldsymbol{A} \cdot \frac{d\boldsymbol{r}_s}{dt}\, \boldsymbol{\nabla} \cdot \boldsymbol{G} \ . \tag{A20}$$

In order to apply eqs. (A19) and (A20) to the problem of an oscillating liquid drop, we need only to replace $d\boldsymbol{r}_s/dt$ with the fluid velocity $\boldsymbol{u}$ at the drop's surface.

[4] Note that we could not apply the divergence theorem here if the surface were not closed.

B Virial Theorem

Here we derive a form of the virial theorem for inviscid, incompressible liquid drops. We demonstrate that, in the limit of small oscillations about a static configuration, the virial theorem implies $\langle E_{\rm K}\rangle = \langle E_{\rm S}\rangle$. It follows that the total mechanical energy is $E_{\rm M} = \langle E_{\rm K} + E_{\rm S}\rangle = \langle 2E_{\rm K}\rangle$, as we show explicitly in § 4.4.

We begin by writing the Euler equation:

$$\rho\frac{d\boldsymbol{u}}{dt} = -\boldsymbol{\nabla} p \ . \tag{B1}$$

Take the dot product of both sides with the position vector $\boldsymbol{r}$ and integrate over volume to obtain

$$\int\limits_{V_d(t)} dV \rho\, \boldsymbol{r}\cdot\frac{d\boldsymbol{u}}{dt} = -\int\limits_{V_d(t)} dV \boldsymbol{r}\cdot\boldsymbol{\nabla} p \ . \tag{B2}$$

Now substitute

$$\boldsymbol{r}\cdot\frac{d\boldsymbol{u}}{dt} = \frac{d}{dt}(\boldsymbol{r}\cdot\boldsymbol{u}) - u^2 = \frac{1}{2}\frac{d^2}{dt^2}r^2 \tag{B3}$$

and

$$\boldsymbol{r}\cdot\boldsymbol{\nabla} p = \boldsymbol{\nabla}\cdot(\boldsymbol{r}p) - 3p \ , \tag{B4}$$

so that

$$\frac{1}{2}\int\limits_{V_d(t)} dV\rho\frac{d^2}{dt^2}r^2 - 2E_{\rm K} = -\int\limits_{V_d(t)} dV\boldsymbol{\nabla}\cdot(\boldsymbol{r}p) + 3\int\limits_{V_d(t)} dV p \ . \tag{B5}$$

Applying the divergence theorem to the first term on the right-hand side, we find

$$\frac{1}{2}\int\limits_{V_d(t)} dV\rho\frac{d^2}{dt^2}r^2 - 2E_{\rm K} = -\int\limits_{A_d(t)} dA\,\hat{\boldsymbol{n}}\cdot\boldsymbol{r}\,p + 3\int\limits_{V_d(t)} dV p \ . \tag{B6}$$

In order to deal with the first term on the left-hand side, consider the following integral:

$$\int\limits_{V_d(t)} dV\frac{dg}{dt} \ . \tag{B7}$$

where g is some scalar function. After substituting $dg/dt = \partial_t g + \boldsymbol{u}\cdot\boldsymbol{\nabla} g$, rewriting $\boldsymbol{u}\cdot\boldsymbol{\nabla} g = \boldsymbol{\nabla}\cdot(\boldsymbol{u}g) - g\boldsymbol{\nabla}\cdot\boldsymbol{u}$, and using eq. (A19), we see that

$$\int\limits_{V_d(t)} dV\frac{dg}{dt} = \frac{d}{dt}\int\limits_{V_d(t)} dV g - \int\limits_{A_d(t)} dA\,\hat{\boldsymbol{n}}\cdot\boldsymbol{u}\,g + \int\limits_{V_d(t)} dV\boldsymbol{\nabla}\cdot(\boldsymbol{u}g) + \int\limits_{V_d(t)} dV g\boldsymbol{\nabla}\cdot\boldsymbol{u} \ . \tag{B8}$$

The last term above vanishes for an incompressible fluid. When we apply the divergence theorem to the third on the right-hand side, the result cancels the second term, and we are left with

$$\int\limits_{V_d(t)} dV\frac{dg}{dt} = \frac{d}{dt}\int\limits_{V_d(t)} dV g \ . \tag{B9}$$

With this result in hand, we can pull the second time derivative outside the first integral in eq. (B6), which gives

$$\frac{1}{2}\frac{d^2}{dt^2}\int\limits_{V_d(t)} dV\rho r^2 - 2E_{\rm K} = -\int\limits_{A_d(t)} dA\,\hat{\boldsymbol{n}}\cdot\boldsymbol{r}\,p + 3\int\limits_{V_d(t)} dV p \ . \tag{B10}$$

Recall from basic mechanics that the moment of inertial tensor is

$$I_{ij} = \int dV \rho (r^2 \delta_{ij} - x_i x_j) \ , \tag{B11}$$

for which the trace is

$$\mathrm{Tr}\{I_{ij}\} = I_{ii} = 2 \int dV \rho r^2 \ . \tag{B12}$$

Let $J \equiv \frac{1}{4} dI_{ii}/dt$, so that

$$\frac{1}{2} \frac{d^2}{dt^2} \int\limits_{V_d(t)} dV \rho r^2 = \frac{dJ}{dt} \ . \tag{B13}$$

In the surface integral of eq. (B10), we make the replacements $\boldsymbol{r} = r_s \hat{\boldsymbol{r}}$ and $p = \sigma \boldsymbol{\nabla} \cdot \hat{\boldsymbol{n}}$ (eq. [73] with $\tau_{ij} = 0$) to obtain

$$\frac{dJ}{dt} - 2E_{\mathrm{K}} = -\sigma \int\limits_{A_d(t)} dA \, r_s \hat{\boldsymbol{n}} \cdot \hat{\boldsymbol{r}} \, \boldsymbol{\nabla} \cdot \hat{\boldsymbol{n}} + 3 \int\limits_{V_d(t)} dV p \ . \tag{B14}$$

Notice that, up to this point, the only important assumptions we have made are that the fluid is incompressible and inviscid. We have said nothing about the geometry, the amplitude of the oscillations, or the nature of the reference equilibrium state.

To derive an approximation of eq. (B14) that is appropriate for small oscillations, we must keep careful track of all $\mathcal{O}(\eta^2)$ terms, since we know that the kinetic and surface energies are $\mathcal{O}(\eta^2)$. Focus first on the surface integral in eq. (B14). For oscillations about a static, spherical equilibrium, we write $\hat{\boldsymbol{n}} = \hat{\boldsymbol{r}} + \delta \hat{\boldsymbol{n}}$, where $\delta \hat{\boldsymbol{n}}$ is the deviation from the equilibrium value. Now, expand $\hat{\boldsymbol{n}}'$ in a series of increasing order in η:

$$\delta \hat{\boldsymbol{n}} = \sum_{i=1} \hat{\boldsymbol{n}}'_i \ , \tag{B15}$$

where the index i gives the order. Given that $\boldsymbol{\nabla} \cdot \hat{\boldsymbol{r}} = 2/r$, the integrand of the surface term in eq. (B14) becomes

$$\begin{aligned} r_s \hat{\boldsymbol{n}} \cdot \hat{\boldsymbol{r}} \, \boldsymbol{\nabla} \cdot \hat{\boldsymbol{n}} &= 2 + (2\hat{\boldsymbol{r}} + r_s \boldsymbol{\nabla}) \cdot \hat{\boldsymbol{n}}' + r_s \hat{\boldsymbol{r}} \cdot \hat{\boldsymbol{n}}' \, \boldsymbol{\nabla} \cdot \hat{\boldsymbol{n}}' \\ &= 2 + [2\hat{\boldsymbol{r}} + a_0(1+\eta)\boldsymbol{\nabla}] \cdot \hat{\boldsymbol{n}}'_1 + (2\hat{\boldsymbol{r}} + a_0 \boldsymbol{\nabla}) \cdot \hat{\boldsymbol{n}}'_2 + a_0 \, \hat{\boldsymbol{r}} \cdot \hat{\boldsymbol{n}}'_1 \, \boldsymbol{\nabla} \cdot \hat{\boldsymbol{n}}'_1 + \dots \ . \end{aligned} \tag{B16}$$

In the second line above, we used $r_s = a(1+\eta)$ with $a = a_0 + \mathcal{O}(\eta^2)$ (see eq. [22]). To alleviate some of the bookkeeping, we point out that $\hat{\boldsymbol{r}} \cdot \hat{\boldsymbol{n}}'_1 = 0$; in the limit $|\hat{\boldsymbol{n}}'| \to 0$, $\hat{\boldsymbol{r}} \cdot \hat{\boldsymbol{n}}'_1 = \hat{\boldsymbol{n}} \cdot \hat{\boldsymbol{n}}' = \frac{1}{2} \delta(\hat{\boldsymbol{n}} \cdot \hat{\boldsymbol{n}}) = 0$, since $\hat{\boldsymbol{n}}$ is a unit vector. Now, to second order,

$$r_s \hat{\boldsymbol{n}} \cdot \hat{\boldsymbol{r}} \, \boldsymbol{\nabla} \cdot \hat{\boldsymbol{n}} = 2 + a_0(1+\eta) \boldsymbol{\nabla} \cdot \hat{\boldsymbol{n}}'_1 + (2\hat{\boldsymbol{r}} + a_0 \boldsymbol{\nabla}) \cdot \hat{\boldsymbol{n}}'_2 \ . \tag{B17}$$

In the static state, the Euler equation tells us that $\boldsymbol{\nabla} p_0 = 0$, and thus, inside the drop, p_0 is a constant equal to the surface pressure $2\sigma/a_0$.

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